

intelligent answers to free-format questions like, "What are the state-of-the-art search engines today?" WebFountain President Robert Carlson says, "WebFountain begins to associate names and people in articles about a certain subject and gathers specific information that can be used as a business intelligence tool." It aims at dealing with large volumes of text, which it collects, stores and analyses. It performs data mining on the data it collects, discovering trends and patterns within.

The three primary components of WebFountain are the Platform, the Data and the Multi-Disciplinary Text Analytics. The first of these, the Platform, is an infrastructure that incorporates various technologies in order to provide complete customer solutions: spiders, miners and other focussed applications. The Data includes data from the internet, newspapers, magazines, blogs and so on. And the third component—the Analytics—consists of technologies and techniques such as NLP, statistical approaches, probability-based approaches, machine learning, AI, etc.

The search bandwagon

Microsoft's Vice President in charge of the Windows division, Jim Allchin, said, "Google's a very nice system, but compared to my vision, it's pathetic." These words signalled Microsoft's entry into the search engine market. Longhorn hopes to search documents irrespective of their location. Microsoft's MSN portal currently uses the Inktomi search technology, but Microsoft recently

Google Alternatives

A crucial aspect of search is the presentation of the results. The current breed of search engines—Google and AltaVista—list results by ranking; but this gives no additional information about each Web site, except a short description. It's easier to make a selection if there's a visual representation of the results. For example, the search engine KartOO (www.kartoo.com), displays results with arcs linking to similar topics, keywords, etc.

Another search engine that does things differently is Vivisimo (www.vivisimo.com); it presents the results in groups, called clusters. For example, if you try a search on 'cricket,' the clusters that show up would be 'world cup,' 'test,' 'sports,' 'cricket club,' and so on. This helps you zoom in onto your desired result site easily.

Teoma (www.teoma.com) is a search engine with a difference—it applies a

measure called 'authority' to determine the relevance of a site. To determine the level of authority of a particular site, Teoma employs proprietary techniques called Refine and Results. Refine refers to Teoma's organising of sites into the communities that occur on the Web—for example, Web sites about music downloads, or those that contain technical papers. These different communities are presented in the browser's results page under Refine. From here, the user can choose the line of search to go along, greatly helping with ambiguity in searches. Next, Results refers to the analysis of the pages, ranking them based on the number of sites that link to it. This is different from Google's approach; where Google takes into consideration all sites that link to a page, Teoma only considers which sites within a particular category of interest refer to a particular page.

announced plans to develop its own algorithmic search engine. It not just Microsoft that's trying to enter the search business: Yahoo!, for instance, has bought Inktomi, even though it currently uses Google's search engine. Amazon.com is another company that has entered this arena. The Seattle-based Amazon has launched a subsidiary named A9, which concentrates on the development of a search engine tuned for e-commerce. These developments are an indication of how serious the busi-

ness is getting.

The Semantic Web is a vision of Tim Berners-Lee, best known as the inventor of the World Wide Web. It aims at structuring information on the Web so that it can be used much more easily by machines, and has been underway for some time now. If it really takes off, it promises to make a difference in the way we deal with online data exchange and e-commerce. Progress in information retrieval and search is likely to be slow, but it is moving at a steady pace. With more and more companies realising how important the business of search is, we can expect more focus on this area of information technology. Indexing data and providing metadata are keys to reducing the problem of the data deluge. People will realise the importance of metadata and indexing more and more, and it is likely that indexing will become more routine. Data mining—the process of extracting previously unknown, and interesting, patterns and information from masses of data—is likely to become more ubiquitous, as we reach the stage where we have too much information and not enough knowledge. ■

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www.dublincore.org

The official Web site of the Dublin Core Metadata Initiative, which aims at developing standards for online metadata.

www.semanticweb.org

The official Web site of The Semantic Web, an initiative that aims at marking up and linking the Web in such a way that it will be more accessible to machine understanding, among other things.

http://javelina.cet.middlebury.edu/lsa/out/search_engines.htm

A site that explains how search engines work, the intricacies of dealing with unstructured data using taxonomies, latent semantic indexing (LSI), and more.

www.unstruct.org

This Web site is all about unstructured information and its management, including how search engines do it, news about projects such as WebFountain, and so on.

<http://tbray.org/ongoing/What/Technology/Search/>

The Search section of the Weblog of Tim Bray, co-creator of XML. Discussed here are several aspects of the search problem.

www.almaden.ibm.com/WebFountain

The homepage of IBM's Almaden Research Center's WebFountain project. WebFountain is a project that aims at going beyond keyword search, bringing intelligence and context into the picture.